

The following problem involves two numbers,  $a_2$  and  $a_3$ , which are given by computer codes that represents them as computable numbers.

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$f(n)$  starts with  $72^{(-C)} (C = 10)$ , and then for  $m = 1, 2, 3, \dots$  adds  $(1/2)^k (k = 72^{(-D \cdot k)})$ , where  $D = 10^{23}$  and  $k = m+2$ .

Thus  $f(n) = 72^{(-C)} + \sum_{k=3}^{n+2} (1/2)^k - \sum_{k=3}^{n+2} 72^{(-D \cdot k)}$ .

As  $n \rightarrow \infty$ :

$$\sum (1/2)^k = 1/4,$$

$$\sum 72^{(-D \cdot k)} = 72^{(-3D)} / (1 - 72^{(-D)}).$$

$$\text{Therefore } a_2 = 1 + 2 \cdot 72^{(-C)} - 2 \cdot 72^{(-3D)} / (1 - 72^{(-D)}).$$

Since  $C = 10 \cdot D$ ,  $2 \cdot 72^{(-C)}$  is much smaller than the subtracted term, making  $a_2 < 1$  (or if the constants were meant to be equal,  $a_2 > 1$ ). In any case,  $a_2 \neq 1$ .

**Values:**

$$a_1 = 1$$

$$a_2 \approx 1 - 2 \cdot 72^{(-3 \cdot 10^{23})} + \dots \text{ (exactly as above, strictly not 1)}$$

$$a_3 = 1$$

**Are they all equal?** No,  $a_1 = a_3 \neq a_2$ .